



Tutorial 2

1. Using Boundary Value Analysis and Equivalence Partitioning test design techniques, identify the optimum set of test cases for testing the business requirements given below.

A system that is designed to work out the tax to be paid should check the following requirements;

- If the salary of an employee is less than Rs. 50,000, then he/she is not taxed.
 - If the salary of an employee is greater than or equal to Rs. 50,000, and less than Rs.200,000, then he/she is taxed at 10%.
 - If the salary of an employee is greater than or equal to Rs. 200,000 and less than Rs. 400,000 then he/she is taxed at 15%.
 - If the salary of an employee is greater than or equal to Rs.400,000, then he/she is taxed at 25%.
2. Create a decision table and the optimized set of combinations of the conditions that could be used for testing the business requirements given below. **Assume that customers are either new or existing customers only.**
Conditions:
 - Existing customers receive a 2% discount on all orders.
 - If it is an existing customer and if he/she pays by cash an additional 3% discount is offered.
 - If fifty or more items are ordered by any type of customer, then in addition to the above-mentioned discounted, the ordered items will be delivered to that customer free of charge.

[Tutorial 2.pptx](#)

Answer

Test Input	Expected output
Equivalence Partitioning	
Salary = 25000	Tax =0
Salary = 100000	Tax = 10%
Salary = 300000	Tax = 15%
Salary = 500000	Tax = 25%
Boundary Value Analysis	
Salary =49999	Tax = 0



Salary = 50000	Tax = 10%
Salary = 50001	Tax = 10%
Salary = 199999	Tax = 10%
200000	Tax = 15%
200001	Tax = 15%
399999	Tax = 15%
400000	25%
400001	25%

2.

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
Existing Customer (2%)	T	T	T	T	F	F	F	F
Pay by cash (3%)	T	T	F	F	T	T	F	F
More than 50 items	T	F	T	F	T	F	T	F
Actions/Outcomes								
Discount	5%	5%	2%	2%	0%	0%	0%	0%
Free Delivery	Y	N	Y	N	Y	N	Y	N